

MAT133: Calculus and Linear Algebra for Commerce

Project 1 First Draft Cover Page

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What is your pod's name?

B-T-W

What is your research question?

Does Tesla's stock price react faster or slower than its deliveries grow?

In your data set, what are the input and output variables? State the quantities in words, including units.

Input: Time • Measured in months (Stock price) • Measured in quarters (deliveries)	Output: Tesla's Stock Price • Measured in USD per share Tesla's Vehicle Deliveries • Vehicles delivered per quarter
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Our project marking team will provide you with feedback on the format, labels, and readability of your scatterplot. In addition to this, you may request feedback on **one** other element of your report, by selecting **one** option below:

- Research question
- Suitability of Dataset
- **Interpretations for the rates of change *****
- Formatting of calculations and mathematical symbols
- Citations

Math 133 Project Report

1. Introduction

Research Question

Does Tesla's stock price react faster or slower than its deliveries grow?

Background

How Tesla shares fell after deliveries dropped 8.5% from a year ago[1]. Total 2024 Q1 deliveries: 386,810 compared to Total 2023 Q1 deliveries: 422,875. This led to TSLA's stock price falling by about 4.9% on April 2, 2024. This delivery decline was due to production disruptions, shipping disruptions caused by the Red Sea conflict, and an arson attack that caused the Berlin factory to be shut down. Weaker demand and increasing competition from Chinese EV companies, were both major factors as well.

The Data Set

Two datasets are used to compare how Tesla's stock price reacts with its vehicle deliveries from January 2023 to December 2025. The first dataset contains TSLA's monthly closing stock prices, measured in USD per share, using the closing price of the asset on the last trading day of each month. The second dataset contains the number of vehicles Tesla delivered to customers worldwide during each quarter, measured in vehicles delivered per quarter. Table 1 shows the first ten months of the monthly stock price data, while table 2 shows the first ten quarters of the quarterly vehicle deliveries data.

The input variable for both datasets is time. For the stock price data, time is measured in months starting from January 2023. For the quarterly deliveries data, time is measured in quarters starting from Q1 in 2023. The output variables are Tesla's monthly closing stock price, measured in USD. And Tesla's quarterly vehicle deliveries, measured in total vehicles delivered to customers each quarter. Yahoo finance[2] is used to obtain the stock price data which records historical market prices and also Tesla's Investor Relations reports[3], where Tesla publishes its quarterly production and delivery results.

These two datasets allow us to directly compare what investors are willing to pay for a TSLA share compared to what the company is actually delivering to customers. I chose the 2023-2025 period because it includes several significant changes in both datasets, including large increases and decreases in stock price and quarterly deliveries.

Data set	Source	Coverage and notes
TSLA monthly closing prices	Yahoo Finance — TSLA Historical Data	Jan 2023-Dec 2025. Closing prices are split-adjusted; values shown in USD.
Tesla quarterly deliveries	Tesla Investor Relations	Q1 2023-Q4 2025. Total vehicle deliveries, not production.

Table 1

TSLA Monthly Closing Price, January–September 2023

Month	Date	Closing price (\$)
1	Jan 2023	173.22
2	Feb 2023	205.71
3	Mar 2023	207.46
4	Apr 2023	164.31
5	May 2023	203.93
6	Jun 2023	261.77
7	Jul 2023	267.43
8	Aug 2023	258.08
9	Sep 2023	250.22
10	Oct 2023	200.07

Table 2

Tesla Quarterly Vehicle Deliveries, Q1 2023–Q2 2025

Quarter	Period	Deliveries (number of vehicles)
1	Q1 2023	422,875
2	Q2 2023	466,140
3	Q3 2023	435,059
4	Q4 2023	484,507

5	Q1 2024	386,810
6	Q2 2024	443,956
7	Q3 2024	462,890
8	Q4 2024	495,570
9	Q1 2025	336,681
10	Q2 2025	384,122

2. Visual Representation of the Data

Figure 1 representing Tesla's monthly closing prices shows a clear uptrend but highly choppy trend within the time period. The largest early increase occurs between months 4 and 6, where the price rose from \$164.31 to \$261.77, almost a \$100 increase, followed by a noticeable decline from month 9 to 10 of around \$50. The most meaningful increase occurs between months 20 to 24 where price rose from \$214.11 to \$403.84, although a large drop occurs directly after during months 25 to 27. Again there is another massive movement upwards in price from months 32 to 33, before settling at that price level. During specific periods I mentioned, the steep rises suggest that the stock price was moving more quickly than quarterly deliberations, which may indicate that there were other factors in investors' expectations during those times.

TSLA monthly closing price, Jan 2023 - Dec 2025

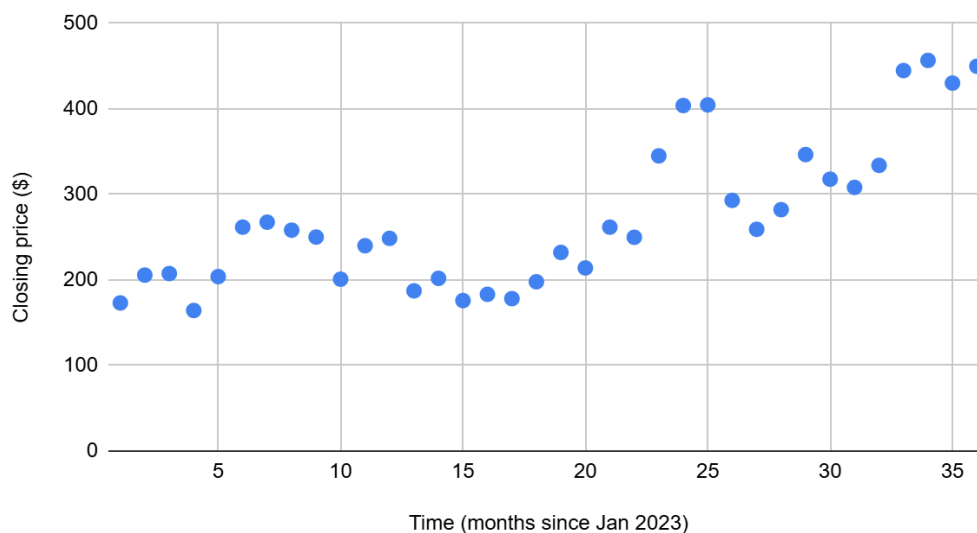


Figure 1: Scatterplot of TSLA Monthly Closing Price

Figure 2 representing Tesla's quarterly deliveries shows a more choppy pattern with no clear overall upward or downward trend. Deliveries rose from 423,000 to 466,000 in quarter 3, fell slightly in quarter 3, and then increased to about 485,000 in quarter 4. The largest declines occur at quarter 5, when deliveries fell to roughly 387,000 and during quarter 9, when they dropped from 495,000 to 336,000. In between those quarters there was a steady increase in overall deliveries. Deliveries then rapidly increased during quarter 11, reaching nearly 497,000 vehicles, followed by a pullback to 418,000. The steep increase from quarter 9 to 11 shows that the rate of change in deliveries was much faster during this period than during any other point in time from 2023-2025. Compared with the stock price graph, the delivery data has no obvious upward trend, which suggests Tesla's stock price was reacting to external factors beyond its current vehicle delivery performance.

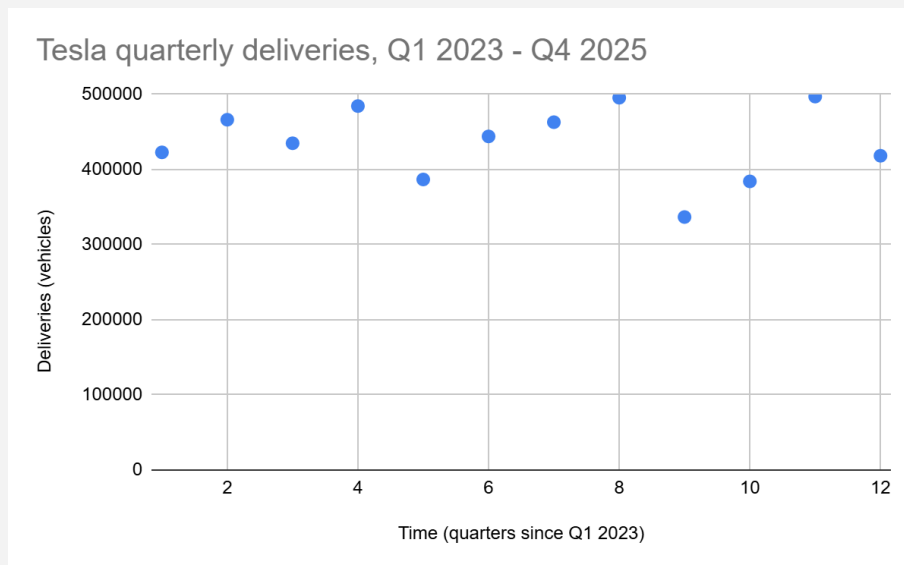


Figure 2: Scatterplot of Tesla Quarterly Deliveries

Overall, the two graphs demonstrate that the largest movements in stock price and vehicle deliveries don't always occur at the same time. This difference supports my research question because it suggests that the stock price may be changing due to external factors or future projection rather than only deliveries data.

3. Rates of Change and Discussion

Average Rate of Change

To calculate the average rate of change clearly, I will use the entire three year period from January 2023, to December 2025.

For Tesla's stock price, the average rate of change is calculated using the beginning and ending prices.

$$(449.72 - 173.22)/(36-1) = \$7.90 \text{ dollars/month}$$

Tesla's stock price rose by an average rate of about \$7.90 per month over the 2023 to 2025 period.

For vehicle deliveries, the average rate of change is:

$$(418227 - 422875)/(12-1) = -422.55 \text{ vehicles/quarter}$$

Deliveries decreased by an average rate of about 423 vehicles per quarter over the same period.

These averages show that while price increased at a rather steady rate, deliveries were nearly flat to slightly negative, but both these averages hide the large increases and decreases that occurred within the 3 year time span.

Instantaneous Rate of Change

To estimate the instantaneous rate of change, I chose to use April 2024 as the specific point because it falls within the Q1 2024 period discussed in the CNBC article.

For Tesla's Stock price, the instantaneous rate of change is:

$$(178.08 - 183.28)/1 = -5.20 \text{ dollars/month}$$

At month 16 (April 2024), Tesla's stock price was falling at about \$5.20 per month based on the following month's price.

For Tesla's quarterly deliveries, the instantaneous rate of change is:

$$(386810 - 484507)/1 = -97697 \text{ vehicles/quarter}$$

Tesla deliveries fell by 97,697 vehicles from Q4 2023 to Q1 2024, which directly matches the disappointing Q1 result discussed in the article.

Relative Rate of Change

For Tesla's stock price in April 2024, the estimated rate of change was $(178.08 - 183.28)/1 = -5.20$ dollars per month, and the April 2024 stock price was \$183.28, so the relative rate was $= -5.20/183.28 \times 100 = -2.84\%$. This means Tesla's stock price was decreasing at a rate of about 2.84% of its April value per month based on the change to May.

For vehicle deliveries, Tesla delivered 386,810 vehicles in Q1 2024 compared with 484,507 in Q4 2023, giving a change of $(386810 - 484507)/1 = -97697$ vehicles/quarter. Compared with the Q1 2024 delivery level, the relative rate was $-97,697/386,810 \times 100 = -25.26\%$. per quarter, meaning the delivery decline was much larger relative to the number of vehicles being delivered than the stock price decline was relative to the share price.

Estimating the Derivative

The derivative was estimated by calculating the rate of change between each pair of consecutive data points. The approximate rates of change were calculated using difference quotients from the data in tables 3 and 4. For each row, I used the right hand difference by subtracting the current value from the following one and dividing by the change in time. For example, the Tesla stock price increased from \$173.22 in January 2023 to \$205.71 in February 2023, giving an approx rate of change of \$32.49 per month. For the final row, a left hand difference was used because there was no following data point. The same method was used for vehicle deliveries, with the time difference adjusted to three months because deliveries were reported quarterly. Figures 3 and 4 were then created using those calculated rates of change, allowing the graphs to show when each function was increasing or decreasing and how rapidly the change was.

Table 3

Estimated Rates of Change of TSLA Closing Price, January–September 2023

Month	Date	Closing price (\$)	Rate of change (\$/month)
1	Jan 2023	173.22	32.49
2	Feb 2023	205.71	1.75
3	Mar 2023	207.46	-43.15

4	Apr 2023	164.31	39.62
5	May 2023	203.93	57.84
6	Jun 2023	261.77	5.66
7	Jul 2023	267.43	−9.35
8	Aug 2023	258.08	−7.86
9	Sep 2023	250.22	−49.38
10	Oct 2023	200.07	−50.15

Note. Each rate is estimated using a right-hand difference: (next period's value – this period's value) ÷ 1 period.

Table 4

Estimated Rates of Change of Tesla Quarterly Deliveries, Q1 2023–Q2 2025

Quarter	Period	Deliveries (vehicles)	Rate of change (vehicles/quarter)
1	Q1 2023	422,875	43,265
2	Q2 2023	466,140	−31,081
3	Q3 2023	435,059	49,448
4	Q4 2023	484,507	−97,697
5	Q1 2024	386,810	57,146
6	Q2 2024	443,956	18,934
7	Q3 2024	462,890	32,680
8	Q4 2024	495,570	−158,889
9	Q1 2025	336,681	47,441
10	Q2 2025	384,122	112,977

Note. Each rate is estimated using a right-hand difference: (next period's value – this period's value) ÷ 1 period.

Figure 3 shows Tesla's monthly price rate of change fluctuates. The largest increase occurs around month 32 at about \$110 per month, while the largest decrease occurs around month 25 at about -\$110 per month.

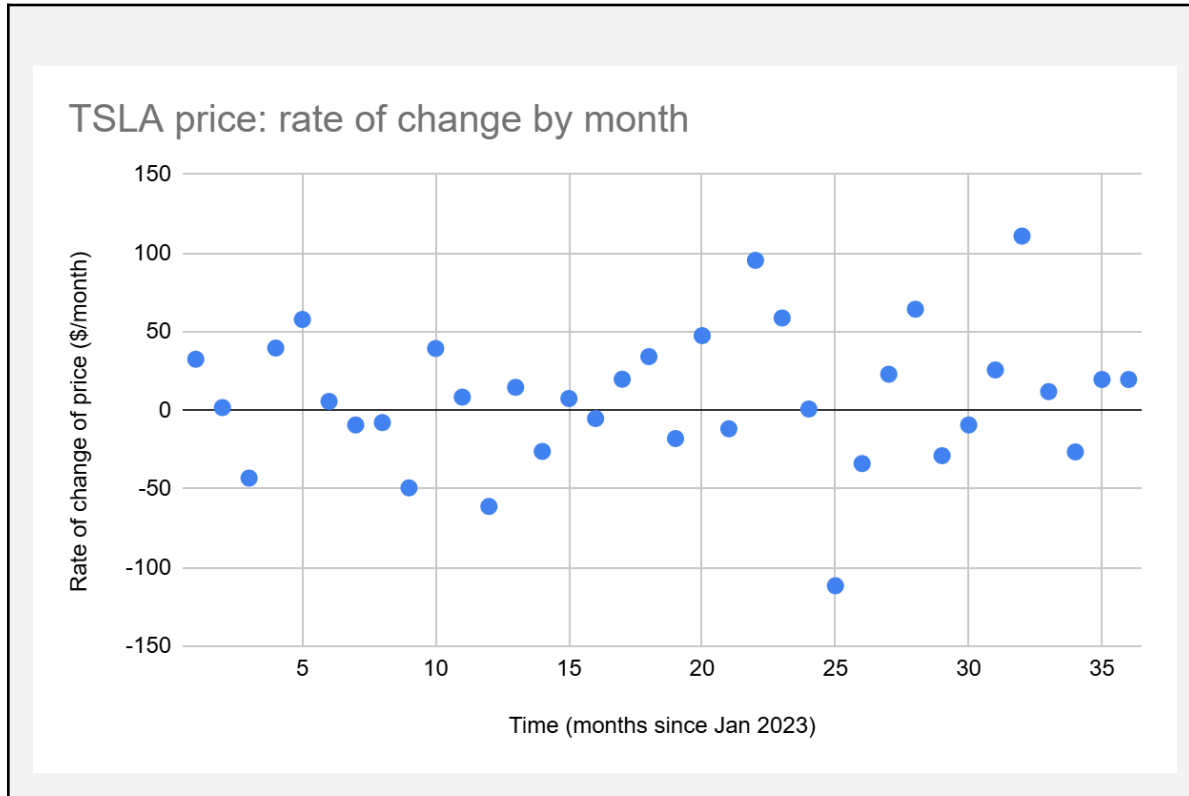


Figure 3: Scatterplot of rate of change of price vs Time in months

Figure 4 shows Tesla's quarterly delivery rate of change over time. The largest increase occurs at quarter 10 at about 110,000 vehicles per quarter, and the largest decrease occurs at quarter 8 at about -160,000 vehicles per quarter.

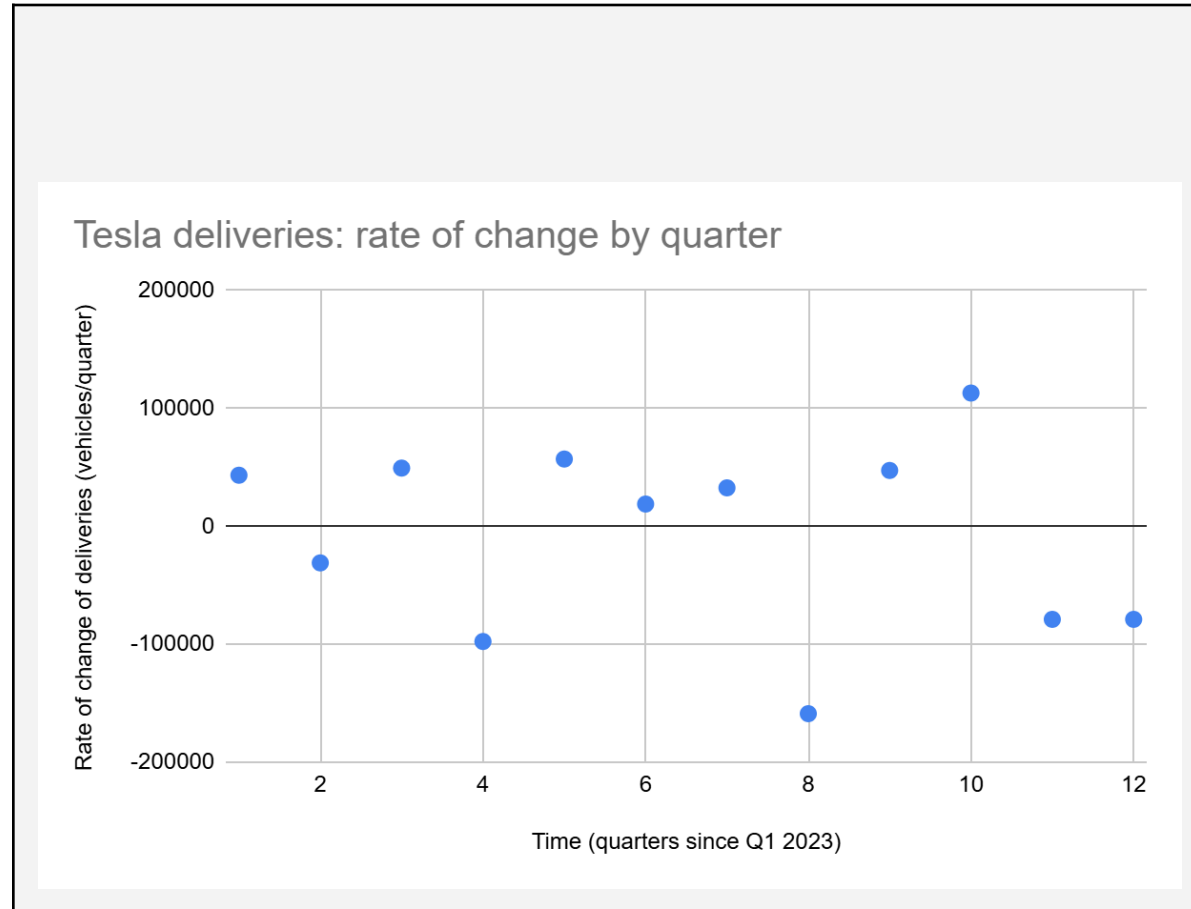


Figure 4: Scatterplot of Rate of change of deliveries vs Time in quarters

Discussion

The rates of change show that Tesla's stock price was much more variable than its vehicle deliveries over the period studied. The stock price frequently switches between positive and negative rates, with some large changes such as the increase from May to June 2023, as well as the large increase near the end of 2024. While vehicle deliveries changed less frequently because they were reported quarterly as opposed to monthly. This is relevant in the real world because a stock price can move quickly due to news and investor expectations, while vehicle deliveries measure actuals over a long period. These factors connect to my research question, "Does Tesla's stock price react faster or slower than its deliveries growth?", because the derivative graphs suggest that the stock price is more frequently variable. Overall, the results suggest that Tesla's stock price is influenced not only by its current deliveries, but many external factors, such as investors' expectations regarding the future of Tesla.

4. Conclusion

Summary of Findings

Overall, the results show that Tesla's stock price changes much more rapidly and frequently than its vehicle deliveries. As shown in the graphs, the stock price had many noticeable increases and decreases throughout 2023-2025 while maintaining an upward trend, while deliveries moved less frequently with no clear trend. This answers the research question by suggesting that Tesla's stock price reacts faster than deliveries growth because investors are always active in the market and can respond immediately to new information, while deliveries take time to change and are reported much less often. With more data, I would investigate whether major changes in Tesla's stock price consistently happen because of specific events, such as earnings reports or production problems, which cause larger or smaller changes than deliveries growth. An upcoming MAT133 topic, such as integration, could help me study accumulated change from rates of change. A real-world event I would investigate is how investor expectations and company news affect stock prices. These would help explain not only how quickly Tesla's stock and deliveries change, but also why they change.

Further Questions and Connections

How do major company events affect Tesla's stock price?

How would integration help measure Tesla's total change over time?

Would using daily stock prices instead of monthly prices change the results?

Does Tesla's stock price change before or after quarterly deliveries are reported?

References

[1] Kolodny, L. (2024, April 2). *Tesla shares fall after deliveries drop 8.5% from a year ago*. CNBC.

[2] Yahoo Finance. (n.d.). *Tesla, Inc. (TSLA) stock historical prices & data*. Yahoo Finance. TSLA Historical Data

[3] Tesla, Inc. (n.d.). *Investor relations*. Tesla. Tesla Investor Relations